



Rethinking XML-enabled agile supply chains

Hyung Jun Ahn^a, Paul Childerhouse^b, Gottfried Vossen^c, Habin Lee^{d,*}

^a College of Business Administration, Hongik University, 94, Wausan-ro, Mapo-gu, Seoul 121-791, Republic of Korea

^b Waikato Management School, University of Waikato, Hamilton 3240, New Zealand

^c Dept. of Information Systems, School of Business Administration and Economics, University of Muenster, 48149 Muenster, Germany

^d Brunel Business School, Brunel University, Kingston Lane, Uxbridge, Middlesex UB8 3PH, UK

ARTICLE INFO

Article history:

Available online 16 July 2011

Keywords:

Agile supply chain
Supply chain
XML
Business to business
ebXML
RosettaNet
Web Services

ABSTRACT

This paper critically discusses the use of Extensible Markup Language (XML) for agile supply chains and proposes practical guidelines and future research directions for the field. XML has been identified by many as a prominent technology for enabling electronic supply chain integration, but the adoption of it in the industry has been rather slow and limited. This paper discusses the factors that enable agility in supply chains, and explains why XML is the best candidate for facilitating such factors. Then the major XML-based integration initiatives such as ebXML, RosettaNet, and Web Services are diagnosed in terms of their pros and cons for supply chain agility, as well as the reasons for their slow diffusion. Based on the discussion, this paper proposes that: the integration efforts need to first focus on the industries that need agility; stringent business process standards are necessary; and implementation should be made much easier and more affordable for small and medium enterprises (SMEs). Further research issues are also suggested based on the discussion.

© 2011 Elsevier Ltd. All rights reserved.

1. Introduction

The term *supply chain management* seems to be everywhere at the moment (Davenport & Brooks, 2004; Hassini, 2008). During the last decade, businesses have seized upon this concept to increase competitiveness. If asked what technology would be most promising for revolutionizing traditional supply chains, many would recommend Extensible Markup Language (XML) which allows seamless exchange of data among heterogeneous information systems of collaborating organizations (Nurmilaakso & Kotinurmi, 2004; Power, 2005).

However, it appears that the adoption of the major XML-based integration initiatives such as ebXML (Naujok & Huemer, 2008), RosettaNet (Boh, Soh, & Yeo, 2007), and Web services (Yan & Klein, 2007) has been rather slow or limited contrary to the expectations (Kanakamedala, King, & Ramsdell, 2003; Liegl, Zapletal, Pichler, & Strommer, 2010; Naujok & Huemer, 2008; Rawlins, 2006; Wilson, 2000). The implementation of them has been mostly by large organizations or their major partners, and the threshold remains still high for small and medium enterprises (SMEs) (Nurmilaakso, 2009; Power, 2008).

This article introduces *agile supply chain management*, arguably one of the most advanced forms of supply chain management (Lee,

2004), where such technological initiatives can make the greatest contribution (Christopher, 2000). By discussing the possible benefits and the current limitations of the technologies in the context of agile supply chains, this article explores future development and research directions for the technology so that its potential capabilities can better enable the next generation of agile supply chains.

This article is organized as follows. Sections 2 and 3 introduce the concept of supply chain agility and a sample case. Section 4 introduces XML and the major integration initiatives based on XML. Section 5 presents the main issues and suggestions for the future development. Section 6 concludes with future research directions.

2. Supply chain agility

The term *supply chain management* was first coined in the early 1980s but supply chains themselves have existed for as long as man has produced and traded goods. They are quite simply the chain of people or organizations that transform earth's resources into consumer products and services. The management of this transformation process is, however, not simple but in many cases involves multiple organizations from multiple countries and contains multiple production and distribution points around the globe. The objective of supply chain management is to perform this transformation process as efficiently as possible and deliver what the end consumer wants as effectively as possible (Christopher, 2011).

Supply chain management really came to the forefront in the 1990s when it was originally seen as an approach to gain efficient-

* Corresponding author. Tel.: +44 18952 67414.

E-mail address: Habin.Lee@brunel.ac.uk (H. Lee).

Table 1
Functional vs. fashionable products (Fisher, 2003).

	Functional	Fashionable
Product life cycle	Long: 2–20+ years	Short: 2–6 months (1 season)
Product variety	Low: 1–5	High: 100s+
Forecasting error	Low: 10%	High: 50–200%
End of season markdown	Rare: 1–5%	Commonplace: 10–25%
Major supply chain costs	Physical: production, delivery and storage	Marketability: obsolescence and stock out
Order winner	Price	Availability
Supply chain strategy	Lean	Agile

cies in ever-tightening markets. Aligned to this has been the uptake of *lean thinking* which originated from Toyota's Just-In-Time (JIT) production that mainly emphasizes removing non-value adding *muda* (a Japanese word for waste), achieving efficiency and low price (Womack & Jones, 2003). There is, however, a practical limit to the leanness when the market is volatile and meeting short time windows is more critical. A typical example is the carpet industry (Johansson, McHugh, Pendlebury, & Wheeler, 1993), where the traditional supply chain required about 16 weeks from raw material supply to the carpet being fitted in the consumers' residence. Although the lean principle can reduce this cycle time to 4 weeks, it hardly helps carpet manufacturers operate in the much more responsive mode needed to satisfy the consumer-imposed time window which can be as short as 1 week. In order to achieve this, production processes were totally re-engineered to be *agile* so that manufacturing was completed in just 3 days through the invention of a new process called Solution Dried Nylon. Table 1 further highlights this need to match supply chain strategy to market requirements.

Functional products have relatively stable demand and long product life cycles. For functional products, consumers do not require excessive choice but they rather purchase primarily based on price. As a result, the costs for this type of supply chain are mainly the production and delivery of the product to the marketplace. In these circumstances, the lean strategy is most appropriate as the focus is on efficiency. At the other extreme are the fashionable products that are short-lived, often only popular for a single season, and are inherently difficult to forecast. Consumers want choice, customization, and the latest version, and hence the major supply chain costs relate to servicing this volatile market and also to obsolescence and stockout risks. A lean supply chain would under-

serve this market, and this is where agility is required with quick turn around times and flexible processes.

In order to achieve agility and thrive in volatile markets, supply chains need strategies and practices that can be characterized by *alignment* and *flexibility* in both physical collaboration and information systems (Fig. 1).

First, alignment is needed so that a whole supply chain can operate in harmony quickly responding to the changes in the market. A close eye needs to be kept on the market so, when the fashion changes, the news is instantly utilized by decision makers to adjust plans and move with the times. Suppliers and manufacturers often need to co-design products from the very beginning to enable flexible manufacturing and logistics. Information systems play a critical role in this alignment where supply chain-wide information should be shared for demand, inventory, and production information, enabling supply chains to advance from being forecast-driven with often distorted and delayed information to market-driven based upon shared real-time demand.

Next, flexibility is needed so that supply chains can react and reconfigure quickly to changing market signals. On the physical side, manufacturers often need the capability to switch between different types of products. Companies should also utilize postponement where the final stage of manufacturing is delayed to meet the latest market demands. A well-known example is Dell where they design common main boards so that they can be packaged for many different options meeting different demands. Companies also need to form flexible partnerships quickly, reconfiguring the supply network for different products and demand characteristics. Information systems again play a very important role, especially for the flexible formation of electronic networks in accordance with physical reconfigurations. The establishment of electronic partnership should be much quicker than, for example, traditional Electronic Data Interchange (EDI) systems which may take months or even years for implementation.

In short, the material flow of agile supply chains is both flexible and responsive to real time volatile markets. Advanced information systems can enable these supply chains to quickly match supply and demand from their virtual networks.

3. A case – complexity in real world supply chains

Many of today's supply chains are under increasing pressure to become agile, making supply chain management more complex accordingly. This can be illustrated through a real-world case. The

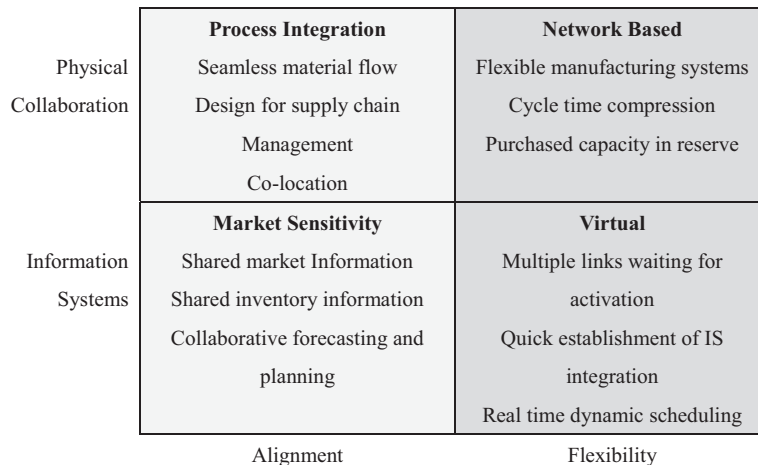


Fig. 1. Factors that make a supply chain agile.

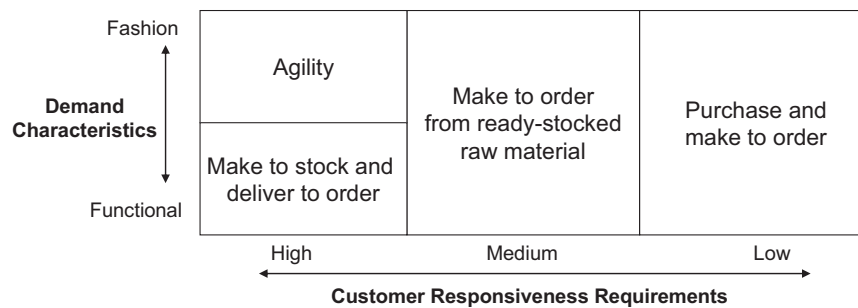


Fig. 2. Matching demand and supply for the global food company of the sample case.

case was developed using Quick Scan (Naim, Childerhouse, Disney, & Towill, 2002) which is a proven method of auditing the health of supply chains. The supply chain is of a Fortune 500 global company producing condiments, sauces, meals, and snacks, holding top market positions in more than 50 countries. The company has so far focused on traditional, staple foods with stable demand over an extended period. As a result, the company could remain competitive by mainly focusing on leanness – efficiency and low price. However, the global food producer is now moving into more fashionable product types such as ready-made meals and new types of condiments, which have volatile demand characteristics, unlike traditional staple foods. This has led to a more diverse range of products which requires a mix of strategies to match supply and demand as depicted in Fig. 2.

The customer responsiveness requirements on the horizontal axis represent how short the lead time needs to be to meet customer expectation. The demand characteristics have been divided into functional versus fashionable on the vertical axis. For products with low lead time requirements, there is enough time for the company to purchase, make, and deliver to specific orders, even if the product is fashionable. If the customer responsiveness requirements are reduced to a medium level, the supply chain only has time to make and deliver from previously purchased raw materials. When customer lead times are short, for the functional products, the company can produce products to forecast as efficiently as possible and deliver to the market on request. This, however, is not appropriate for the more fashionable prod-

ucts because the stockpile of finished products is very risky as the demand may not materialize or, alternatively, the demand may boom and result in a stock-out situation with large numbers of unsatisfied customers. It is in this situation when agility is most needed.

This case suggests two important points. First, as companies innovate with new products and globalize into new markets, there is increased uncertainty and volatility that needs to be managed by agile practices. Second, many companies often produce a range of products that needs multiple supply chain strategies together such as leanness, agility, or hybrids. The complexity can further increase when there are a large number of heterogeneous trading partners with different relationship requirements.

This complexity often requires the use of multiple heterogeneous inter-organizational information systems. Fig. 3 provides an intuitive understanding for this where different types of e-partnering options are located along the two dimensions of agility, alignment and flexibility. We can see that, for example, traditional or Internet-based EDI can be used for transaction-oriented and long-term stable relationships, while tightly integrated virtual networks can be used for highly aligned and flexible relationships. It is obvious that the options located at the upper-right corner of Fig. 3 can better serve agile supply chains, while the systems located on the left hand side of the figure often can be adopted by lean supply chains. Hence, ideally, many companies in the supply chains, such as that of the case study, require a mix of multiple online partnering

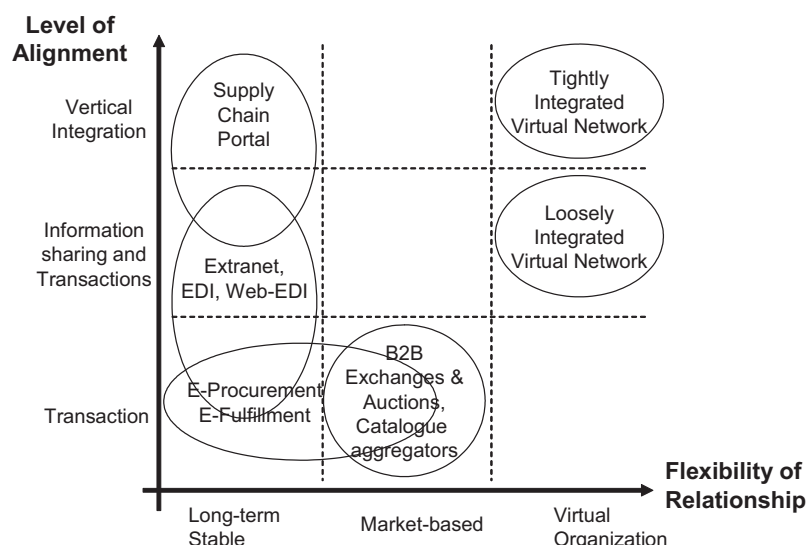


Fig. 3. Alternative online collaboration scenarios (see Mahadevan, 2003 for brief introduction of the e-partnering options in the diagram).

Table 2
Comparison of major XML-based initiatives for supply chain integration – compiled from various sources (Alonso et al., 2004; Damodaran, 2004; Gibb & Damodaran, 2002).

	ebXML	RosettaNet	Web services
1. Brief history	Initiated in 1999. First suite of specifications released in May 2001.	Founded in 1998. First Partnership Interface Process (PIP) implementation in 1999.	Key specifications such as Simple Object Access Protocol (SOAP), Universal Description, Discovery and Integration (UDDI), and Web Services Description Language (WSDL) first released around 1998–2002.
2. Industry focus	Cross-industry.	Focusing on electronic components, IT, and semiconductor manufacturing industries so far.	Cross-industry.
3. Standard bodies	OASIS and UN/CEFACT	RosettaNet	Mainly World Wide Web Consortium (W3C), OASIS, and Web Services Interoperability (WS-I).
4. Key specifications	ebXML Collaborative Partner Profile Agreement (CPPA), ebXML Messaging Service (MS), ebXML Registry Information Model (RIM), ebXML Registry Services (RS), etc.	Multiple Messaging Services (MMS), PIPs, RosettaNet Implementation Framework (RNIF), Dictionaries, etc.	SOAP, WSDL, UDDI, WS-Coordination, WS-Transaction, Web Services Flow Language (WSFL), Business Process Execution Language (BPEL), Web Services Choreography Description Language (WS-CDL), etc.
5. Standard business processes and vocabulary	Cross-industry specifications that include Catalog of Common Business Processes and Core Components Dictionary.	PIPs and Dictionaries for specific business processes and vocabulary.	BPEL for specifying executable business processes. Specific business processes need to be developed by industries and users.
6. Adoption by SMEs for supply chain integration	No strong evidence of wide adoption by SMEs. Adoption cases are mainly limited to public sector organizations such as National Health Service (NHS) in UK or Centers for Disease Control and Prevention (CDC) in US.	Relatively successful in the focus area. However, no strong evidence of wide adoption by SMEs. Implementation cases mainly limited to large companies and their key partners.	Despite its popularity as a general Service-Oriented Architecture (SOA) framework, no strong evidence of SMEs widely adopting it for supply chain integration.
7. Key challenges	Slow adoption. Implementation still expensive for SMEs. Lack of specific standard business processes for implementation.	Slow adoption. Implementation still expensive for SMEs. Case-by-case and lengthy implementations.	More popular as a general SOA framework. Specific business processes need to be developed for industries. Overhead in protocol stack.

options to support different strategies with many trading partners and products. However, this is impractical for many SMEs in the supply chain.

4. XML – a key enabler of supply chain integration

XML, the *Extensible Mark-up Language*, is a World Wide Web Consortium (W3C) recommendation for a general-purpose mark-up language that supports a wide variety of applications, in particular in information integration scenarios, and allows users to introduce application-specific tags and define new mark-up languages. The basic strength of XML for business-to-business (B2B) communication comes from its extensibility, Internet-based openness, and machine-readable structure (Nurmilaakso & Kotinurmi, 2004). The extensibility of XML allows the modeling and creation of virtually any business document or process; the openness of XML enables inexpensive exchange of XML-based data on the Internet; and its well-defined structure makes the interpretation of XML documents by computer programs easy. These properties are all essential for seamless B2B integration that requires automated transactions and information sharing.

XML, however, is a low-level technology that needs more elaboration at a higher level for actual supply chain integration. For this high-level business integration, there have been many approaches to establishing standards for business documents, business processes, and related services. Among them, the three most widely known are the ebXML and RosettaNet initiatives as well as the development of Web services (see Table 2 for summary) (Nurmilaakso & Kotinurmi, 2004; Papazoglou & Ribbers, 2006). ebXML is a joint effort by Organization for the Advancement of Structured Information Standards (OASIS) and The United Nations Centre for Trade Facilitation and Electronic Business (UN/CEFACT), and focuses on the development of cross-industry infrastructure and specifications for supply chain integration (Gibb & Damodaran,

2002). RosettaNet, initiated by a consortium of industry members and technology vendors, is focused on creating specifications for business documents and processes for specific areas such as electronic components and semiconductor manufacturing industries (Damodaran, 2004). Web services enable businesses to communicate and integrate through invoking services of each other based on the concept of Service Orientation (Alonso, Casati, Kuno, & Machiraju, 2004). Web services are supported by a suite of specifications for message formats, service interfaces, and business processes.

Although there are differences among the initiatives, they all have strong potential for supply chain integration, especially in comparison with EDI. Together with the basic strengths of XML, the initiatives are aimed at enabling low cost and flexible implementation of online partnerships based on the standards and infrastructure. Different from the simple transactions and document exchanges typical of older technologies, the initiatives can also support sophisticated forms of supply chain coordination such as collaborative forecasting and collaborative design. They also provide directory services for online partnering options so that companies can quickly search for and establish flexible relationships. Overall, all the above provide strong advantages suitable for enabling the alignment and flexibility much needed by agile supply chains.

5. Suggestions for future development

In spite of many advantages, the adoption of the major XML-based integration initiatives so far has been quite limited to large companies and their partners in specific industry sectors, public sector organizations, or deployment of simple services that are far from being business process-level integration (Table 2) (Naujok & Huemer, 2008; Nurmilaakso, 2008, 2009; Rawlins, 2006; Yan & Klein, 2007). Although it might be too early to judge their success

Table 3

Factors affecting the success of Internet-based B2B integration (Chatterjee et al., 2006; Dai & Kauffman, 2002).

Category	Factors
1. Technology issues	Technical complexity Lack of standards and use of proprietary formats Integration with other information systems Security High cost of implementation and maintenance
2. Shared vision for benefits	Lack of benefit sharing models; exploitation by strong players Threat to existing supply chain relationships Fitting market characteristics Trust in partnership
3. Organizational issues	Willingness to change Top management support for changes

or failure for supply chain integration at this stage, this adoption pattern is very different from what has been expected of them. Therefore, considering that it has already been more than 10 years since the inception of the initiatives, it is time that we raised concerns about the adoption and suggested a direction for their future development.

Apart from some case studies highlighting mainly successful aspects of implementations, rigorous empirical investigation on the factors of their adoption is yet to be conducted. Nonetheless, we can utilize abundant studies on the adoption of earlier Internet-based B2B or EDI systems for the discussion of the challenges. Table 3 summarizes key factors to the success of B2B systems and Internet-based EDI found from literature categorized into three groups: technology issues, vision for shared benefits, and organizational issues (Chatterjee, Segars, & Watson, 2006; Dai & Kauffman, 2002). Based on the factors in the table combined with our earlier discussion regarding the advancement of agile supply chain management, we next discuss the reasons for their slow adoption status and present development directions for the success of the initiatives in the future.

5.1. Focus on the areas that need agility

As we have seen in Section 2, there are some areas where agility is needed more than others and, hence, will be under greater pressure to adopt the XML-based initiatives. Supply chains for only functional and commodity products will enjoy long term and stable relationships with lean principles, having little incentive for taking the high risks of making huge investment in new technologies, and potentially harming existing relationships with their partners by introducing unnecessary flexibility. For this reason, it is observed that many companies that have been using EDI for a long time are not very willing to switch to XML, especially those with lean supply chains, having no immediate additional benefits (Nurmilaakso, 2008, 2009). In contrast, for sectors operating in volatile markets, e.g. ready-made meals or new condiments exploring new markets in our case study of Section 3, agility will become more and more a matter of survival. Companies operating in the areas will have clearer motivation to employ the technologies that can directly help them to win competition over non-agile rivals. So far, much of the efforts in the initiatives have been led by consortiums of big technology companies or industry organizations without any special focus on agile supply chains (Liegl et al., 2010; Naujok & Huemer, 2008; Yan & Klein, 2007). As such, focusing on those areas that need agility first can help the XML-based initiatives take off successfully, effectively overcoming many of the barriers, such as the organizational issues and lack of shared vision in the second and third categories of Table 3.

5.2. Create stringent business process standards

Another critical issue that impedes the supply chain-wide adoption of the technologies is lack of stringent, vendor-independent business process standards. Without them, implementations can remain costly and incompatible, making a critical barrier to achieving the high level alignment and flexibility for agile supply chains. Although the standards are relatively stringent at the infrastructure-level such as with SOAP and WSDL in the Web Services area, Messaging Services (MS) of ebXML, and RosettaNet Implementation Framework (RNIF) of RosettaNet, there is a serious gap at the higher level for specific business processes that are essential to achieve the plug-and-play style interoperability (see the fifth row of Table 2). Unless the standards are provided for business processes, many heterogeneous processes can be developed for each transaction type, which will inevitably lead to complex and costly implementations and incompatibility, impeding the adoption. The absence of specific process standards also sets the initial threshold high for companies and technology vendors, because they need to make extra investment first for developing the processes.

ebXML has so far mainly focused on infrastructure-level specifications, leaving actual business process specifications to individual implementers, which has led to proprietary and incompatible business processes (Rawlins, 2006). In contrast, RosettaNet does provide specific business process specifications known as Partnership Interface Processes (PIPs) to which RosettaNet's relative success can be largely attributed (Boh et al., 2007; Damodaran, 2004). However, the PIPs are mainly for the electronics and semiconductor industries, and other industries have to newly develop PIPs to use RosettaNet. PIPs are also known to allow too much customizability which can often result in lengthy case-by-case incompatible implementations (Olson & Williams, 2001). Web Services has gained its position as a popular and general Service-Oriented Architecture (SOA) framework. Nevertheless, because Web Services is not generally regarded as a tool specialized for supply chain integration, not much effort has been devoted to the development and standardization of business processes exclusively for Web Services, leaving the responsibility to individual industries and companies adopting it (Papazoglou, Traverso, Dustdar, & Leymann, 2008). Hence, in order to successfully kick start the adoption of the initiatives, it is imperative for the initiatives to work with industries and put more efforts into developing specific and stringent business processes first.

5.3. Make implementation easy for SMEs

In order to achieve the high level of alignment for agility, it is essential that the technologies are accepted not only by large companies and key partners, but also by small partners on the supply chain. Although XML is an open technology working on the Internet, the technologies can still be expensive for SMEs considering the costs of implementation, maintenance, and required IT skill (Harland, Caldwell, Powell, & Zheng, 2007; Nurmilaakso, 2009; Vaaland & Heide, 2007). Furthermore, management of an online relationship for each SME can be very complicated due to a wide variety of partnering options that might be required, a large number of trading partners, product diversity, and strategic diversity, as has been shown in Fig. 3. This often creates unmanageable complexity for SMEs so that it is not surprising to see many companies, including many small partners of the global food company in our earlier case, still relying on email, fax, or telephones for order processing (Harland et al., 2007).

Therefore, the cost of implementation needs to be significantly reduced for SMEs, for which there might be many useful approaches. For example, XML-based business process standards and messaging infrastructure can be embedded in small applica-

tions such as e-fulfillment systems or order processing systems for SMEs. Currently, more efforts are focused on implementing a lower level infrastructure for high-end servers or case-by-case implementations, which is largely due to the lack of stringent specifications of business processes and limited market sizes. Another possible approach is to provide business process integration as a form of online service using the Application Service Providers (ASPs) model or the Software-as-a-Service (SaaS) paradigm (Vossen & Hagemann, 2007) so that SMEs can use them with minimum IT overhead. We can also borrow the concept of *agile software development*, which is widely adopted in the context of Web 2.0 applications, where empowerment of developers and interaction with customers are emphasized over strict and lengthy development process, especially for digital products (Agile Manifesto, 2001; Kappel, Proll, & Reich, 2006; Vossen & Hagemann, 2007). One remarkable program addressing this adoption problem for SMEs is RosettaNet Automated Enablement (RAE) in RosettaNet which aims at providing simplified and affordable solutions to SMEs (RosettaNet, 2007). These approaches, together with stringent business processes, can help SMEs overcome the technology barriers (the first category of Table 3) effectively.

6. Conclusion and future research directions

Consumer's expectations are constantly increasing. In order to keep up with ever-changing market requirements and shortening lead times, more businesses are developing agile supply chains. This new strategy requires live market knowledge and virtual corporations, both of which can only be enabled through advanced information systems. It is here where we see the opportunity for XML-based initiatives to link together all the supply chain members through information enrichment. This article has presented a timely discussion for the further development of the initiatives by identifying their potential for agile supply chains as well as analyzing their current limitations. In line with the discussion, this paper concludes by suggesting a couple of future academic research directions.

First, as much argued in this paper, academic studies are needed along with practical efforts to create affordable and practical standards for SMEs so that the XML-based integration can achieve holistic alignment. The standards need to be simple and robust so that both implementation and maintenance costs can be minimized for SMEs.

Second, another stream of research required is analyzing the expected costs and benefits of adopting XML for various ways of supply chain integration. It was discussed that there are many overlooked obstacles in adopting the technology that may increase costs and even make the adoption unviable. Also, individual implementation choices of XML-based integration can vary resulting in different styles and strengths of partnership, and different levels of complexity in business processes. Therefore, the assessment of the overall costs and benefits can help companies make appropriate design decisions and invest in the technology with more confidence. Simulation is possibly a good method for the analysis considering the complexity of the problem and the diversity of the variables involved (Marquez, 2010).

Third, XML needs to be combined with many intelligent systems technologies. While XML provides a way of seamless data exchange, intelligent systems can further boost the benefits of XML by supporting or automating the complex decision makings required for managing supply chain resources and relationships in an agile way. For example, business intelligence (BI) and data mining can be used to detect changes in the market and help companies proactively adapt to the changes (Watson & Wixom, 2007). Another example is the multi-agent technology which can be used for efficiently

and autonomously managing the distributed and heterogeneous supply chain resources (Ahn & Lee, 2004). That is, by employing the intelligent systems technology and combining it with XML properly, we can expect to decrease the difficulty of XML-based integration, and therefore make it more affordable for agile supply chains.

Although we may still have a rough road ahead, with more attention paid as well to other important issues such as building trust among partners and creating benefit-sharing models, the authors believe the suggestions and discussions in this paper will help the XML-based initiatives become successful enablers of agile supply chains.

References

- Agile Manifesto (2001). Manifesto for Agile Software Development. <http://agilemanifesto.org/>.
- Ahn, H., & Lee, H. (2004). An agent-based dynamic information network for supply chain management. *BT Technology Journal*, 22, 18–27.
- Alonso, G., Casati, F., Kuno, H., & Machiraju, V. (2004). *Web Services: Concepts, architectures, and applications*. Berlin: Springer Verlag.
- Boh, W. F., Soh, C., & Yeo, S. (2007). Standards development and diffusion: A case study of RosettaNet. *Communications of the ACM*, 50, 57–62.
- Chatterjee, D., Segars, A. H., & Watson, R. T. (2006). Realizing the promise of E-Business: Developing and leveraging electronic partnering options. *California Management Review*, 48, 60–83.
- Christopher, M. (2000). The agile supply chain: Competing in volatile markets. *Industrial Marketing Management*, 29, 37–44.
- Christopher, M. (2011). *Logistics and supply chain management* (4th edition). FT Press.
- Dai, Q., & Kauffman, R. J. (2002). B2B E-commerce revisited: Leading perspectives on the key issues and research directions. *Electronic Markets*, 12, 67–83.
- Damodaran, S. (2004). B2B integration over the Internet with XML: RosettaNet successes and challenges. In *13th International World Wide Web conference – Alternate track papers & posters* (pp. 188–195). New York: ACM Press.
- Davenport, T. H., & Brooks, J. D. (2004). Enterprise systems and the supply chain. *Journal of Enterprise Information Management*, 17, 8–19.
- Fisher, M. L. (2003). What is the right supply chain for your product? *Operations Management: Critical Perspectives on Business and Management*, 105, 73.
- Gibb, B. K., & Damodaran, S. (2002). *ebXML: Concepts and application*. New York, NY, USA: John Wiley & Sons, Inc.
- Harland, C. M., Caldwell, N. D., Powell, P., & Zheng, J. (2007). Barriers to supply chain information integration: SMEs adrift of eLands. *Journal of Operations Management*, 25, 1234–1254.
- Hassini, E. (2008). Building competitive enterprises through supply chain management. *Journal of Enterprise Information Management*, 21, 341–344.
- Johansson, H. J., McHugh, P., Pendlebury, A. J., & Wheeler, W. A. (1993). *Business process reengineering: Breakpoint strategies for market dominance*. Chichester: John Wiley & Sons.
- Kanakamedala, K., King, J., & Ramsdell, G. (2003). The truth about XML. *McKinsey Quarterly*, 3, 9–12.
- Kappel, G., Proll, B., & Reich, S. (2006). *Web engineering: The discipline of systematic development of web applications*. New York: John Wiley & Sons.
- Lee, H. (2004). The Triple-A Supply Chain. *Harvard Business Review*, 82, 102–112.
- Liegl, P., Zapletal, M., Pichler, C., & Strommer, M. (2010). State-of-the-art in business document standards. In *8th IEEE international conference on industrial informatics* (pp. 234–241). Osaka, Japan: IEEE.
- Mahadevan, B. (2003). Making sense of emerging market structures in B2B e-commerce. *California Management Review*, 46, 86–100.
- Marquez, A. C. (2010). *Dynamic modelling for supply chain management: Dealing with front-end, back-end and integration issues*. Springer Verlag.
- Naim, M. M., Childerhouse, P., Disney, S. M., & Towill, D. R. (2002). A supply chain diagnostic methodology: Determining the vector of change. *Computers and Industrial Engineering*, 43, 135–157.
- Naujok, K., & Huemer, C. (2008). Designing ebXML – The Work of UN/CEFACT. In *Ontologies-based business integration*. Springer., pp. 79–93.
- Nurmilaakso, J.-M. (2008). EDI, XML and e-business frameworks: A survey. *Computers in Industry*, 59, 370–379.
- Nurmilaakso, J.-M. (2009). EDI-based and XML-based business-to-business integration: A statistical analysis. *International Journal of Business Information Systems*, 4, 639–654.
- Nurmilaakso, J.-M., & Kotinurmi, P. (2004). A review of XML-based supply chain integration. *Production Planning & Control*, 15, 608–621.
- Olson, E. G., & Williams, P. R. (2001). *E-commerce with web-based standards: A case study in implementing RosettaNet process standards at Avnet, Inc.* RosettaNet.
- Papazoglou, M. P., & Ribbers, P. (2006). *e-Business: Organizational and technical foundations*. Chichester: John Wiley & Sons.
- Papazoglou, M. P., Traverso, P., Dustdar, S., & Leymann, F. (2008). Service-oriented computing: A research roadmap. *International Journal of Cooperative Information Systems*, 17, 223–255.
- Power, D. (2005). Supply chain management integration and implementation: A literature review. *Supply Chain Management: An International Journal*, 10, 252–263.

- Power, D. (2008). Adoption of supply chain management enabling technologies: Comparing small, medium and larger organizations. *Operations and Supply Chain Management*, 1, 31–42.
- Rawlins, M. (2006). ebXML – A critical analysis. Last retrieved on March 9, 2011 from: <http://www.rawlinsecconsulting.com/ebXML/>.
- RosettaNet. (2007). RosettaNet automated enablement. Last retrieved on March 9, 2011 from: http://www.rosettanet.org/dnn_rose/tabid/383/Default.aspx.
- Vaaland, T. I., & Heide, M. (2007). Can the SME survive the supply chain challenges? *Supply Chain Management: An International Journal*, 12, 20–31.
- Vossen, G., & Hagemann, S. (2007). *Unleashing Web 2.0: From concepts to creativity*. San Francisco: Morgan Kaufmann.
- Watson, H. J., & Wixom, B. H. (2007). The current state of business intelligence. *Computer*, 40, 96–99.
- Wilson, T. (2000). EDI is alive and kicking, study says. *InternetWeek*, 801, 15.
- Womack, J., & Jones, D. (2003). *Lean thinking: Banish waste and create wealth in your corporation, revised and updated*. Free Press.
- Yan, Y., & Klein, M. (2007). Web Services vs. ebXML: An evaluation of Web Services and ebXML for e-Business applications. In R. G. Qiu (Ed.), *Enterprise service computing: from concept to deployment*. Idea Group Publishing.